

Business Requirements Document

EHR Workflow Redesign

Compliance-driven claims workflow redesign for a leading UAE health insurer

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Regulatory scope	DHA (Dubai) · DoH/HAAD (Abu Dhabi) · Shafafiya · eSMA
Linked projects	P1 Claims Denial · P2 RCM Dashboard · P3 Cost Modelling
Document status	Complete — for sign-off across Provider, Insurer, Regulator
Total gap recovery	AED 9,200,000 across 6 gaps and 3 delivery phases

This document summarises the Confluence BRD and the Jira delivery backlog (6 Epics, 18 user stories) produced for the EHR Workflow Redesign project. All financial figures are derived from the P1 claims dataset (4,500 claims) and the P3 chronic-disease cohort (3,000 members).

1 Executive Summary

The current claims workflow across Provider, Insurer, and Regulator is largely manual, producing a portfolio denial rate of 33.6% and AED 7,550,894 in denied claim value. Analysis across Projects 1 and 3 identified six structural gaps, each carrying a quantified AED impact and a DHA/DoH regulatory consequence.

The redesigned (to-be) workflow reduces 15 manual steps to 10 automated or semi-automated steps, introduces real-time ICD-10 validation, EHR-to-RCM integration, an automated pre-authorisation rules engine, deployment of the P3 predictive model (AUC 0.950) for high-cost member identification, and automated denial routing. Total recoverable value is estimated at **AED 9,200,000**, delivered across three phases over nine months.

Financial Case

Metric	As-Is	To-Be Target	Improvement
Overall denial rate	33.6%	<15%	55% reduction
Inpatient denial rate	32.7%	<12%	63% reduction
ICD-10 coding error rate	35.5% (GP)	<5%	86% reduction
Pre-auth compliance	Inconsistent	99%+	Full compliance
AR days	45+ days	<25 days	44% faster
High-cost members flagged	0% (reactive)	87% (proactive)	New capability

2 Scope

In Scope

Claims workflow from clinical coding through submission, adjudication, denial management, and high-cost member identification across all three parties.

Out of Scope

Clinical treatment decisions, EHR vendor selection, and member-facing portals are outside this redesign and handled under separate initiatives.

Delivery Phases

Phase	Theme	Scope	Months	Recovery
Phase 1	Foundation	ICD-10 validation + EHR-RCM integration	1–3	AED 2,700,000
Phase 2	Automation	Pre-auth + denial follow-up + compliance	4–6	AED 3,100,000
Phase 3	Intelligence	P3 predictive model deployment	7–9	AED 3,400,000

3 Gap Analysis

Six gaps were identified across the three parties. Raw AED impact reflects total financial exposure; estimated recovery reflects the recoverable amount after the solution is implemented. The BRD and delivery phases use recovery figures.

#	Gap	Party	Raw Impact	Recovery	Risk
1	ICD-10 coding errors — no validation	Provider	AED 2,642,813	AED 1,600,000	High
2	No EHR to RCM integration	Provider/Insurer	AED 1,834,887	AED 1,100,000	High
3	Manual pre-authorisation	Insurer	AED 1,834,887	AED 1,200,000	Critical
4	No high-cost member identification	Insurer	AED 6,783,941	AED 3,400,000	High
5	No automated denial follow-up	Insurer	AED 7,550,894	AED 400,000	High
6	No pre-submission compliance check	All parties	AED 1,887,724	AED 1,500,000	Critical

Total estimated recovery: **AED 9,200,000** (figures rounded to nearest AED 100,000). Recovery scoped per gap — see P3 Notebook 04 and P4 Notebook 02 for full derivations.

4 Jira Delivery Backlog

The project is structured in Jira as 6 Epics (one per gap), each broken into 3 user stories — 18 stories total (SCRUM-11 to SCRUM-28). Each epic maps directly to a gap and its required solution.

EP-01 — ICD-10 Real-Time Validation (Gap 1)

Story	User Story
SCRUM-1	As a clinical coder, I want ICD-10 codes validated at point of entry so errors are caught before submission.
SCRUM-2	As a billing lead, I want a rejection-reason dashboard so I can see which codes fail most often.
SCRUM-3	As a compliance officer, I want validation rules updated when DoH/HAAD code sets change.

EP-02 — EHR to RCM Integration (Gap 2)

Story	User Story
SCRUM-1	As a billing analyst, I want claims auto-generated from the EHR so I stop re-keying data.
SCRUM-2	As an IT lead, I want a bi-directional API layer between EHR and RCM systems.
SCRUM-3	As a finance manager, I want claim data to flow automatically on discharge or consultation close.

EP-03 — Automated Pre-Authorisation (Gap 3)

Story	User Story
SCRUM-1	As an intake officer, I want inpatient claims without valid pre-auth flagged and held automatically.
SCRUM-2	As a medical reviewer, I want a rules engine to check pre-auth status at claim intake.
SCRUM-3	As a compliance officer, I want a full audit trail of pre-auth decisions for DHA reporting.

EP-04 — High-Cost Member Identification (Gap 4)

Story	User Story
SCRUM-1	As a care manager, I want members predicted to become high-cost flagged daily by the P3 model.
SCRUM-2	As an analyst, I want model outputs integrated with the care-management referral system.
SCRUM-3	As a medical director, I want intervention outcomes tracked to measure model ROI.

EP-05 — Automated Denial Follow-Up (Gap 5)

Story	User Story
SCRUM-1	As a billing analyst, I want denied claims auto-routed by denial reason to the right team.
SCRUM-2	As an AR lead, I want a 90-day resubmission tracker with alerts before the window closes.
SCRUM-3	As a finance manager, I want denial-recovery rates reported by reason code.

EP-06 — Pre-Submission Compliance Check (Gap 6)

Story	User Story
SCRUM-1	As a submission officer, I want every claim checked against DHA/DoH rules before it leaves the insurer.
SCRUM-2	As a compliance officer, I want ICD-10 validity, pre-auth, eligibility, and timing all verified in real time.
SCRUM-3	As an IT lead, I want DHA/DoH rule updates applied automatically to the compliance engine.

5 Success Metrics

Project success is measured against the as-is baseline using three metric families: claims KPIs (denial rate, AR days), financial recovery (AED realised per phase), and model KPIs (P3 AUC 0.950, 87% high-cost member recall). Targets are listed in the Executive Summary financial case table.

6 Assumptions and Constraints

Assumes continued DHA/DoH regulatory framework (Shafafiya/eSMA), availability of historical claims data for model retraining, and provider/insurer system access for integration. Constrained by the 90-day submission window and mandatory pre-auth for inpatient procedures.

7 Sign-Off

Party	Role	Approval
Provider	Hospital / clinic operations	_____
Insurer	Claims & medical review	_____
Regulator	DHA / DoH compliance liaison	_____

This PDF is a static snapshot of the live Confluence BRD and Jira backlog, committed to GitHub for portfolio accessibility. Live artifacts are maintained in Atlassian Confluence and Jira.